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Assignment 7

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 117 = 3 + 114 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:

3 points: Correct answer with work shown

2 points: Incorrect answer but attempt shows some understanding (work shown)

1 point: Incorrect answer but an attempt was made (work shown)

0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]
(link to your live google doc)

https://docs.google.com/document/d/1dzAgyy7UhyrmR9CyIVDsXdToMj1JATrVwGZaqPQtogo/edit?usp=sharing

I. The Binomial Distribution (10 x 3 = 30 pts)

Directions: For the situations below, define a random variable X for the situation and then decide if they follow a binomial distribution model by commenting on the four requirements.

1) You roll a DnD dice (20-sided), 20 times, and record the number that shows on the dice.

X = number on die
Two possible outcomes: more than 2
Fixed probability: 1/20 per face
Independent trials: rolls are independent
Fixed trials: n = 20

Not binomial because more than two possible outcomes.

2) A basketball player can make 60% of their free throws. The coach plans on having a free throw shooting competition and the player will be shooting 100 shots.

X = number of free throws made
Two possible outcomes: make/miss
Fixed probability: p = 0.6
Independent trials: shots are independent
Fixed trials: n = 100

Binomial distribution model would be appropriate, assuming shots are truly independent.

3) From a standard deck of cards, you pull out a card, record the suit, put it back, and reshuffle. You continue until you get two spades in a row.

X = number of cards drawn
Two possible outcomes: red/black
Fixed probability: p = 0.5
Independent trials: draws are independent
Fixed trials: not fixed number of trials

Not binomial because there isn't a fixed number of trials.

4) You are conducting a survey at your school to see how many students own smartphones. The probability that a student will own a smartphone is 0.85. You plan on having 200 students participate in your survey.

X = number of students with smartphones
Two possible outcomes: owns a smartphone/ does not own a smartphone
Fixed probability: p = 0.85
Independent trials: Assume random sampling
Fixed trials: n = 200

Binomial distribution model would be appropriate.

5) Zach is an 85% field goal kicker. Each goal he kicks is independent of the next kick. At practice today, he will be doing 30 kicks.

For the following table, fill in the missing pieces.

Situation	***Probability Notation***	***Formula***	***Value***
What is the probability Zach will make exactly 25 kicks?	P(X = 25)	$(30\ 25\ )0.\ 85^{25}0.\ 15^{5}$	0.1861
What is the probability Zach will make exactly 20 kicks?	P(X = 20)	$(30\ 20\ )0.\ 85^{20}0.\ 15^{10}$	0.0067

What is the probability Zach will make no more than 20 kicks?	P(X ≤ 20)	P(X = 1) + P(X = 2) + ... + P(X = 19) + P(X = 20)	0.0097
What is the probability Zach will make no more than 24 kicks?	P(X ≤ 24)	P(X = 1) + P(X = 2) + ... + P(X = 23) + P(X = 24)	0.2894
What is the probability Zach will make exactly 28 kicks?	P(X = 28)	$\binom{30}{28} 0.85^{28} 0.15^{2}$	0.1034
What is the probability Zach will make more than 26 kicks?	P(X > 26)	P(X = 27) + P(X = 28) + P(X = 29) + P(X = 30)	0.317

II. The Geometric Distribution (14 x 3 = 42 pts)

1) For the following situations, decide if it is a binomial setting, a geometric setting, or neither. Explain your answer.

a) You keep drawing cards out a deck, without replacement, until an ace is drawn.

Neither, because the probability is not fixed and trials are not independent.

b) You roll a dice 20 times and record the number of sixes you have rolled.

Binomial because the number of trials is fixed, there are two possible outcomes (6/ not 6), the probability is fixed, and the trials are independent.

c) Crest claims that 40% of Americans use their toothpaste. You take a random sample of 50 Americans and count how many use Crest toothpaste.

Binomial, because there are two possible outcomes, with a fixed probability, independent trials, and a set number of trials.

d) You flip a coin until you get tails.

Geometric, because there are two possible outcomes, with a fixed probability, independent trials, and trials are repeated until there is a success.

e) 5% of the tomatoes at a farmer's market have imperfections on them. You randomly choose one tomato at a time until you find one with an imperfection.

Geometric, because there are two possible outcomes, with a fixed probability, independent trials, and trials are repeated until there is a success.

f) 5% of the tomatoes at the farmer's market have imperfections on them. You randomly choose 20 tomatoes and count the number of imperfections on them.

Neither, because the number of possible outcomes is more than two.

2) In a recent Pew study, it was found that 28% of all homes have at least one dog. Let X = the number of houses you look at before finding a house with a dog. Find and interpret the following probabilities.

a) $P(X = 3)$

$P(X = 3) = 0.72^2 0.28 = 0.1452$

The probability of finding a dog at the third house visited is 0.1452.

b) $P(X < 5)$

$P(X < 5) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 0.7313$

The probability of finding at least one dog at the first 4 houses visited is 0.7313.

c) P(X > 4)

P(X > 4) = 1 - (P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)) = 0.2687

The probability of finding no dogs at the first 4 houses visited is 0.2687.

d) E(X)

E(X) = 1/p = 3.571

On average, you need to visit 3.571 houses before finding a house with a dog.

3) Zach's teammate, Taylor, is the quarterback for the football team. She can complete 65% of her passes.

For the following table, fill in the missing pieces.

Situation	***Probability Notation***	***Formula***	***Value***
What is the probability that it takes Taylor 3 incomplete passes before she has a completion?	P(X = 4)	$(0.35)^3(0.65)$	0.0279
What is the probability that it takes Taylor no more than 1 incomplete pass before she has a completion?	P(X < 3)	P(X = 1) + P(X = 2)	0.8775
What is the probability that it takes Taylor 4 incomplete passes before she has a completion?	P(X = 5)	$(0.35)^4(0.65)$	0.0098
What is the probability that it takes Taylor more than 4 incomplete passes before she has a completion?	P(X > 5)	1 – (P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5))	0.0053

III. The Normal Distribution and Combining Normal Random Variables (4 x 3 = 12 pts)

1) Consider a set of 9000 scores on a national test that is known to be approximately normally distributed with a mean of 500 and a standard deviation of 90.

(a) What is the probability that a randomly selected student has a score greater than 600? [Write your numerical answer in the box, and your work of deriving and calculating in the empty space]

Answer: $\boxed{0.1333}$

X = test scores

$X \sim N(500, 90)$

$P(X > 600) = P(z > (600 - 500)/90) = P(z > 1.11) = 0.1333$

(b) How many scores are there between 450 and 600?[Write your numerical answer in the box, and your work of deriving and calculating in the empty space]

Answer:

5197

X = test scores
$X \sim N(500, 90)$
$P(450 < X < 600) = P((450 - 500)/90 < z < (600 - 500)/90) = P(-0.56 < z < 1.11) = 0.5775$

$9000 * 0.5775 = 5197$

(c) Megan needs to be in the top 1% of the scores on this test to qualify for a scholarship. What is the minimum score Megan needs? [Write your numerical answer in the box, and your work of deriving and calculating in the empty space]

Answer:

710

X = test scores
$X \sim N(500, 90)$
$P(X > x) = 0.01$
$P(z > (x - 500)/90) = 0.01$
$z = 2.33$
$2.33 > (x - 500)/90$
$209.7 > x - 500$
$709.7 > x$

2) Gus and Cal go bowling every week. Gus's scores are normally distributed with a mean of 175 pins and a standard deviation of 30 pins. Cal's scores are normally distributed with a mean of 150 pins and a standard deviation of 40 pins. Assume that their scores in any given game are independent. Let G be Gus's score in a random game, C be Cal's score in a random game, and D be the difference between Gus's and Cal's scores where D = G – C. What is the probability that Gus will knock down more pins than Cal? [Write your numerical answer in the box, and your work of deriving and calculating in the empty space]

Answer:

0.6915

$\mu_D = \mu_G - \mu_C = 175 - 150 = 25$

$\sigma_D^2 = \sigma_G^2 + \sigma_C^2 = 90 + 160 = 250$; $\sigma_D = 50$

$P(D > 0) = P(z > (0 - 25)/50) = P(z > -0.5) = 0.6915$

IV. Sampling Distribution of Sample Proportions (8 x 3 = 24 pts)

1) Suppose a large candy machine has 15% orange candies. Imagine taking an SRS of 25 candies from the machine and observing the sample proportion $\hat{p}$ of orange candies. [Write your numerical answer in the box, and your work of deriving and calculating in the empty space]

(a) What is the mean of the sampling distribution of $\hat{p}$? Why?

Answer:

0.15

mean of sampling distribution = population parameter
15% is population parameter so the mean of the sampling distribution will be 0.15

(b) Find the standard deviation of the sampling distribution of $\hat{p}$. Check to see if the 10% condition is met.

Answer:

0.0714

25 < 0.1(all candies in the machine); number of candies is not given but we can assume large means more than 250 candies
σ_p = sqrt(p(1-p)n) = sqrt(0.15*.85/25) = sqrt(0.0051) = 0.0714

(c) Is the sampling distribution of $\hat{p}$ approximately Normal? Check to see if the Large Counts condition is met.

Answer:

no

np = 25*0.15 = 3.75
n(1-p) = 25*0.85 = 21.25

np < 10
n(1-p) > 10

(d) If the sample size were 225 rather than 25, how would this change the sampling distribution of $\hat{p}$?

Answer:

0.02 < 0.07

the mean would stay the same but the spread would decrease

2) The Harvard College Alcohol Study finds that 67% of college students support efforts to "crack down on underage drinking". Does this result hold at a large local college? To find out, college administrators surveyed an SRS of 100 students and found that 62 support a crackdown on underage drinking. What is the probability that the proportion in an SRS of 100 students is 0.62 or less? Does this refute the claim made by Harvard? **Explain using the 4 step process**.

Answer:

Step 1.

State:

p = proportion of students support a crackdown on underage drinking
n = 100
p = 0.67
$P(\hat{p} \leq 0.62)$

Step 2.

Plan:

Random: SRS is random
$\mu_{\hat{p}} = 0.67$
Independent: 100 ≤ 0.1(all students at a large local college)
$\sigma_{\hat{p}}$ = sqrt(0.67*0.23/100) = 0.0393
Normal: large counts condition
$0.67(100) \geq 10$
$0.23(100) \geq 10$
the sampling distribution is approximately normal

Step 3.

Do:

$P(\hat{p} \le 0.62)$ $\hat{p} \sim N(0.67, 0.0393)$
$P(z \le (0.62-0.67)/0.0393) = P(z \le -1.27) = 0.102$

Step 4.

Conclude:

The probability that the proportion in the SRS of 100 students is 0.62 or less is 0.102. Using a significance level of 0.05, this result does not refute the claim by Harvard.

V. Sampling Distribution of Sample Means (2 x 3 = 6 pts)

1) The Wechsler Adult Intelligence Scale (WAIS) is a common "IQ test" for adults. The distribution of WAIS scores for people over 16 years of age is approximately normal with mean 100 and standard deviation 15. What is the probability that the average WAIS score of an SRS of 60 people is 105 or higher? [Write your numerical answer in the box, and your work of deriving and calculating in the empty space]

Answer: 0.2594

$P(x \geq 105)$, $\mu = 100$, $\sigma = 15$, $n = 60$

Random: SRS stated — $\mu_x = 100$

Independent: $60 \leq 0.1$(people over 16) — $\sigma_x = 15/\text{sqrt}(60) = 7.75$

Normal: population is normal

$P(x \geq 105) = P(z \geq (105-100)/7.75) = P(z \geq 0.645) = 0.2594$

3) A company that owns and services a fleet of cars for its sales force has found that the service lifetime of disc brake pads varies from car to car according to a Normal distribution with mean $\mu = 55{,}000$ miles and standard deviation $\sigma = 4500$ miles. The company installs a new brand of brake pads on 8 cars. The average life on the pads on these 8 cars turns out to be 51,800 miles. What is the probability that the sample mean lifetime is 51,800 miles or less if the lifetime distribution is unchanged? (The company takes this probability as evidence that the average lifetime of the new brand of pads is less than 55,000 miles.) [Write your numerical answer in the box, and your work of deriving and calculating in the empty space]

Answer: 0.0221

$P(x \leq 51800)$, $\mu = 55000$, $\sigma = 4500$, $n = 8$

Random: assume pads on 8 new cards are randomly selected from the new brand — $\mu_x = 55000$

Independent: $8 \leq 0.1$(all pads manufactured by new brand) — $\sigma_x = 4500/\text{sqrt}(8) = 1591$

Normal: population distribution is normal

$P(x \leq 51800) = P(z \leq (51800-55000)/1591) = P(z \leq -2.01) = 0.0221$

The probability is so low that this would be taken as evidence that the average lifetime of the new brand of pads is less than 55,000 miles.

THE END